A
Project Report
On
SportiQ – Smart Sports Equipment Management & Borrowing Mobile Application

Prepared by

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IT368 – Project - III
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DEPARTMENT OF INFORMATION TECHNOLOGY

Faculty of Technology & Engineering, CHARUSAT

Devang Patel Institute of Advance Technology and Research (DEPSTAR)

At: Changa, Dist: Anand – 388421

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CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CERTIFICATE

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DECLARATION BY THE KARTIK (23DIT015)/VRAJ (23DIT056)/ANSH (23DIT071)/MAULYA (23DIT072)

We hereby declare that the project report entitled "SportiQ – Smart Sports Equipment Management & Borrowing Mobile Application" submitted by us to Devang Patel Institute of Advance Technology and Research (DEPSTAR), Changa in partial fulfilment of the requirements for the award of the degree of **B.Tech Information Technology**, from the Information Technology, DEPSTAR, FTE is a record of bonafide IT368 Project-III carried out by us under the guidance of Prof. Sachin Patel. We further declare that the work carried out and documented in this project report has not been submitted anywhere else either in part or in full and it is the original work, for the award of any other degree or diploma in this institute or any other institute or university.

Signature of the candidate
(Kartik Guleria – 23DIT015)

Signature of the candidate
(Vraj Patel – 23DIT056)

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

TABLE OF CONTENTS

ABSTRACT

SportiQ is a mobile-based application developed to digitize and streamline the management of sports equipment in educational institutions such as colleges and universities. Traditional systems rely on manual record-keeping using registers or spreadsheets, which leads to inefficiencies such as lack of real-time availability, difficulty in tracking borrowed items, delayed returns, and lack of accountability.

The proposed system provides a centralized and user-friendly solution where students can view available equipment, borrow items using QR or barcode scanning, track due dates, and monitor penalties for late returns. Administrators can efficiently manage inventory, monitor borrowing activities, and generate reports through a dedicated admin interface.

The application is developed using modern technologies including Flutter for frontend development, Node.js and Express.js for backend services, MongoDB for database management, and Firebase Authentication for secure login. The system ensures real-time data updates, improved transparency, and reduced manual workload.

SportiQ aims to provide a scalable, efficient, and cost-effective solution that enhances resource utilization and improves the overall experience for both students and administrators.

CHAPTER 1: INTRODUCTION

1.1 BACKGROUND

In many educational institutions, sports equipment management is handled using traditional manual methods such as paper registers or spreadsheets. These systems are time-consuming, prone to human errors, and lack real-time tracking capabilities. As a result, it becomes difficult for administrators to maintain accurate records of equipment availability, borrowing, and return status.

With the advancement of mobile and cloud technologies, there is a need to replace these outdated systems with a digital solution that ensures efficiency, transparency, and ease of access. A mobile-based application can provide real-time updates, better tracking, and improved management of resources.

1.2 PROBLEM STATEMENT

The existing manual system for managing sports equipment suffers from several limitations:

- Lack of real-time information about equipment availability
- Difficulty in tracking borrowed and returned items
- Delays in returning equipment due to lack of monitoring
- Manual calculation of penalties leading to errors
- Increased administrative workload
- Risk of equipment loss or misuse

These issues highlight the need for a reliable and automated system that can efficiently manage sports equipment.

1.3 OBJECTIVES

The main objectives of the SportiQ system are:

- To digitize sports equipment management
- To provide real-time tracking of equipment availability
- To simplify the borrowing and return process
- To reduce manual workload and errors
- To ensure accountability through automated tracking and penalties
- To develop a scalable and user-friendly mobile application

1.4 SCOPE OF THE PROJECT

The scope of the project includes the development of a mobile-based system that allows students and administrators to manage sports equipment efficiently. Students can view available items, borrow equipment using QR or barcode scanning, track their usage, and view penalties. Administrators can manage inventory, monitor activities, and generate reports.

The system is designed for use within educational institutions and can be extended in the future to support multiple campuses and advanced analytics features.

CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION

Literature review involves studying existing systems, research papers, and solutions related to the project domain. It helps in understanding the strengths and limitations of current approaches and provides insights for designing an improved system.

2.2 EXISTING SYSTEMS

Several systems have been developed for asset and inventory management; however, most of them are either manual or designed for large-scale enterprises. In educational institutions, sports equipment is commonly managed using paper registers or basic spreadsheets.

These systems suffer from multiple drawbacks such as lack of real-time updates, difficulty in tracking usage, and dependency on manual calculations. Additionally, they do not provide automated features such as penalty calculation or digital tracking of borrowed items.

2.3 RELATED WORK

Some modern inventory management systems use digital platforms and databases to track assets. However, these systems are often complex, costly, and not specifically tailored for educational environments.

Mobile-based solutions with QR or barcode scanning have been used in other domains to improve tracking and automation. These approaches demonstrate the effectiveness of digital systems in reducing manual effort and improving accuracy.

2.4 LIMITATIONS OF EXISTING SYSTEMS

The major limitations observed in existing systems include:

- Lack of real-time data availability
- High dependency on manual record-keeping
- No proper tracking of borrowed and returned items
- Absence of automated penalty calculation
- Limited accessibility and scalability

2.5 PROPOSED SYSTEM ADVANTAGES

The proposed system, SportiQ, overcomes these limitations by providing:

- Real-time equipment tracking
- Automated borrow and return workflow
- QR/barcode-based identification
- Role-based access for students and administrators
- Automated penalty calculation
- Scalable and user-friendly interface

CHAPTER 3: SYSTEM ANALYSIS

3.1 INTRODUCTION

System analysis involves studying the existing system, identifying its limitations, and defining the requirements for the proposed system. It helps in understanding what needs to be improved and how the new system will address existing problems.

3.2 EXISTING SYSTEM

The current system for managing sports equipment in many educational institutions is manual. Records are maintained using paper registers or spreadsheets.

Limitations of the existing system:

- No real-time information about equipment availability
- Difficulty in tracking borrowed and returned items
- High chances of human errors
- Manual calculation of penalties
- Increased administrative workload
- Lack of transparency and accountability

3.3 PROPOSED SYSTEM

The proposed system, SportiQ, is a mobile-based application that automates sports equipment management. It provides a centralized platform for students and administrators to manage equipment efficiently.

Features of the proposed system:

- Real-time tracking of equipment
- QR/barcode-based borrowing system
- Automated penalty calculation
- Role-based access (Student and Admin)
- Easy monitoring and reporting

3.4 FUNCTIONAL REQUIREMENTS

The system should be able to perform the following functions:

Student Module:

- Login using secure authentication

- View available equipment
- Borrow equipment using QR/barcode
- Track borrowed items and due dates
- View penalty details

Admin Module:

- Manage equipment inventory (Add/Update/Delete)
- Monitor borrowing and return activities
- Track overdue items
- Generate reports

3.5 NON-FUNCTIONAL REQUIREMENTS

- **Performance:** Fast response time and real-time updates
- **Security:** Secure authentication and role-based access
- **Scalability:** Ability to handle increasing users and data
- **Usability:** User-friendly mobile interface
- **Reliability:** Accurate data handling and minimal errors

3.6 FEASIBILITY STUDY

- **Technical Feasibility:**
  Uses modern and widely supported technologies like Flutter, Node.js, and MongoDB.
- **Economic Feasibility:**
  The system is built using free and open-source tools, making it cost-effective.
- **Operational Feasibility:**
  The system is easy to use and requires minimal training.

CHAPTER 4: TECHNOLOGY STACK

4.1 INTRODUCTION

Technology stack refers to the set of tools, frameworks, and technologies used to develop and deploy the application. The selection of appropriate technologies is important to ensure performance, scalability, and ease of development.

4.2 FRONTEND TECHNOLOGY

Flutter (Dart):

Flutter is used for developing the mobile application. It provides a fast and responsive user interface with a single codebase for multiple platforms. Flutter allows building visually attractive and dynamic applications.

4.3 BACKEND TECHNOLOGY

Node.js:

Node.js is used to build the backend server. It enables fast and scalable server-side development.

Express.js:

Express.js is a lightweight framework used with Node.js to create RESTful APIs and handle server-side logic efficiently.

4.4 DATABASE

MongoDB:

MongoDB is a NoSQL database used to store application data. It provides flexibility in schema design and supports scalable data storage.

Mongoose:

Mongoose is used as an Object Data Modeling (ODM) library to define schemas, validations, and relationships in MongoDB.

4.5 AUTHENTICATION

Firebase Authentication:

Firebase Authentication is used to provide secure login functionality using Google Sign-In. It ensures user identity verification and secure access control.

4.6 TOOLS USED

- **Figma :** Used for designing wireframes and UI layouts
- **Draw.io:** Used for creating system architecture and diagrams
- **VS Code:** Used for coding and development
- **Postman:** Used for API testing

CHAPTER 5: SYSTEM DESIGN

5.1 INTRODUCTION

System design defines the architecture, components, and data flow of the application. It provides a structured approach to implement the system efficiently. This chapter focuses on the overall architecture and database design of the SportiQ system.

5.2 SYSTEM ARCHITECTURE

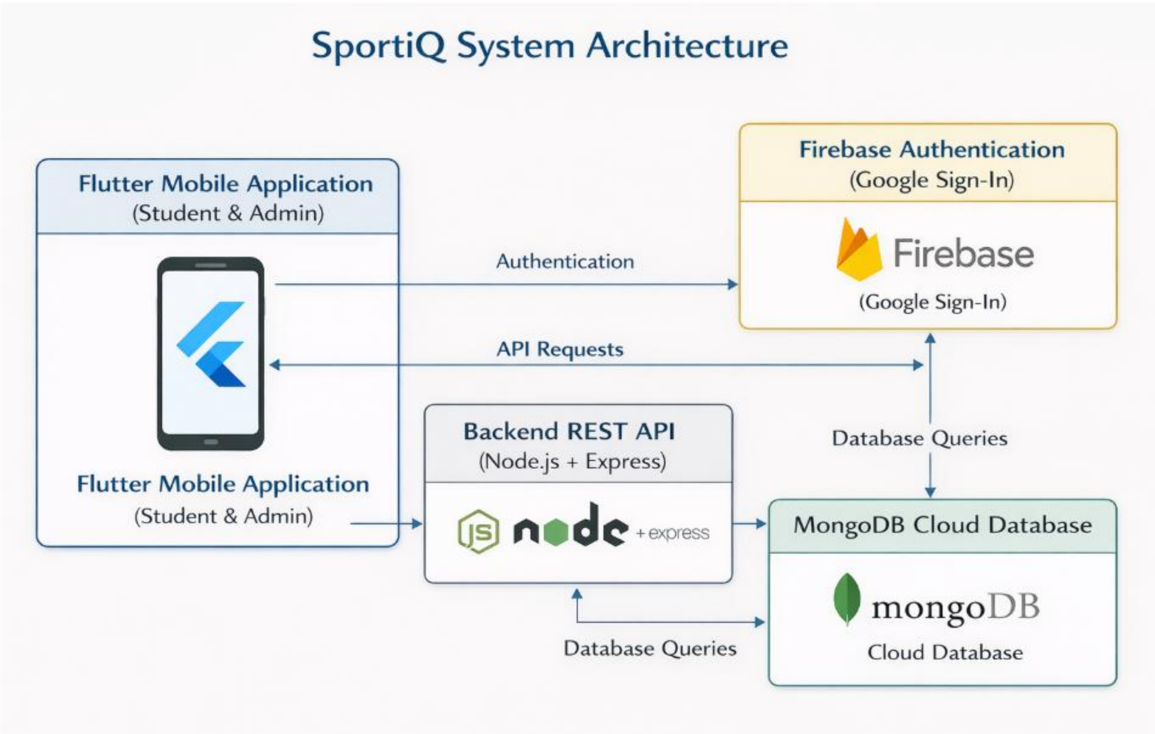
The SportiQ system follows a client-server architecture where the mobile application communicates with the backend server through APIs.

Architecture Flow:

Flutter Mobile App → Node.js (Express API) → MongoDB Database → Firebase Authentication

- The Flutter app handles user interaction
- The Node.js backend processes requests and business logic
- MongoDB stores application data
- Firebase Authentication manages user login and security

☞ *System Architecture Diagram*



5.3 DATABASE DESIGN

The database is designed using MongoDB with a collection-based structure to ensure flexibility and scalability.

Main Collections:

1. **User Collection** : Stores user details such as name, phone number, college ID, and role (student/admin)
2. **Equipment Collection** : Stores equipment details such as name, type, barcode ID, total quantity, and available quantity
3. **Rental Collection** : Stores borrowing details such as user reference, equipment reference, borrow time, return status, and timestamps

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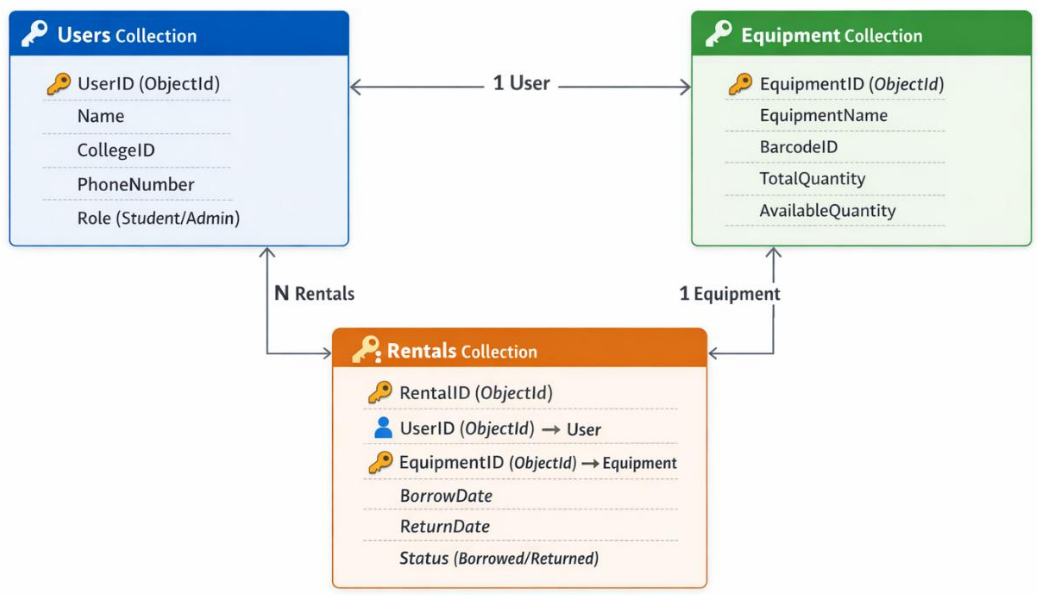
5.4 RELATIONSHIP BETWEEN COLLECTIONS

- **User ↔ Rental:** One user can have multiple rental records
- **Equipment ↔ Rental:** One equipment can be borrowed multiple times

Relationships are implemented using **ObjectId references** to maintain data consistency and enable efficient querying.

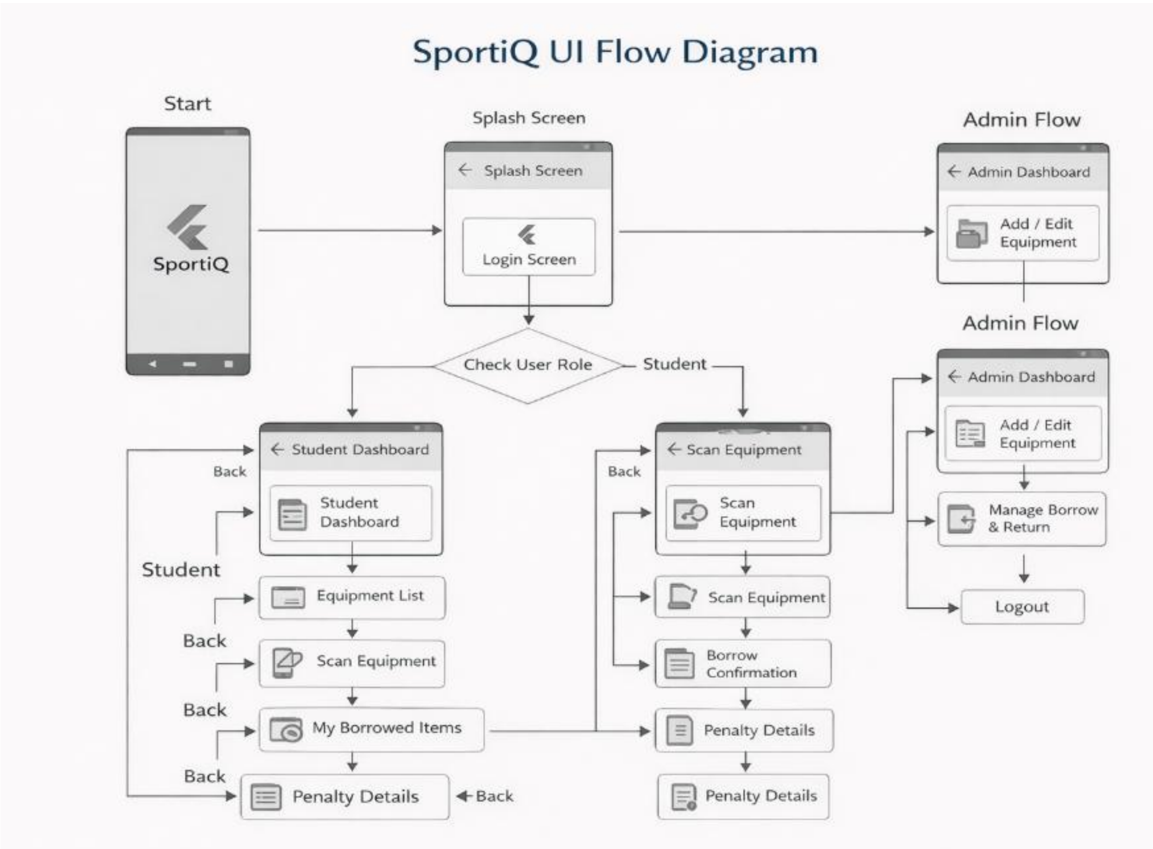
☞ *(Insert ER Diagram Here)*



5.5 DATA FLOW (BORROW & RETURN PROCESS)

1. Student logs into the system
2. Scans QR/barcode of equipment
3. Request is sent to backend API
4. Database checks equipment availability
5. Rental record is created
6. Equipment quantity is updated
7. On return, status is updated and penalty is calculated (if applicable)

☞ *Workflow Diagram*



5.6 DESIGN FEATURES

- Scalable and flexible database structure
- Efficient data relationships using references
- Real-time updates of equipment availability
- Separation of concerns between frontend, backend, and database

CHAPTER 6: TESTING

6.1 INTRODUCTION

Testing is an essential phase in software development that ensures the system works correctly and meets the specified requirements. It helps in identifying errors, validating functionality, and improving the overall quality of the system.

6.2 TESTING OBJECTIVES

The main objectives of testing in the SportiQ system are:

- To verify correct implementation of all modules
- To ensure accurate data handling and storage
- To validate the borrow and return workflow
- To check system reliability and performance
- To detect and fix errors

6.3 TYPES OF TESTING

6.3.1 UNIT TESTING

Individual components such as database models and API endpoints were tested independently to ensure correctness.

6.3.2 INTEGRATION TESTING

The interaction between frontend, backend, and database was tested to ensure smooth data flow.

6.3.3 SYSTEM TESTING

The complete system was tested as a whole to verify overall functionality.

6.4 TEST CASES

Test Case ID	Description	Expected Result	Actual Result
TC-01	User Login	User successfully logs in	Passed
TC-02	View Equipment	Equipment list displayed	Passed
TC-03	Borrow Equipment	Rental record created	Passed
TC-04	Return Equipment	Status updated correctly	Passed

Test Case ID	Description	Expected Result	Actual Result
TC-05	Penalty Calculation	Correct penalty shown	Passed

6.5 RESULTS OF TESTING

- All major functionalities were tested successfully
- Borrow and return workflow worked correctly
- Data consistency was maintained across collections
- Minor issues were identified and resolved

6.6 LIMITATIONS OF TESTING

- Limited testing under high user load
- Some edge cases may require further validation

CHAPTER 7: RESULTS

7.1 INTRODUCTION

This chapter presents the outcomes obtained after implementing and testing the SportiQ system. The results demonstrate how effectively the system meets the project objectives and solves the identified problems.

7.2 SYSTEM OUTPUT

The SportiQ application successfully provides a digital platform for managing sports equipment. The system allows users to perform operations such as login, viewing equipment, borrowing items, returning items, and checking penalties.

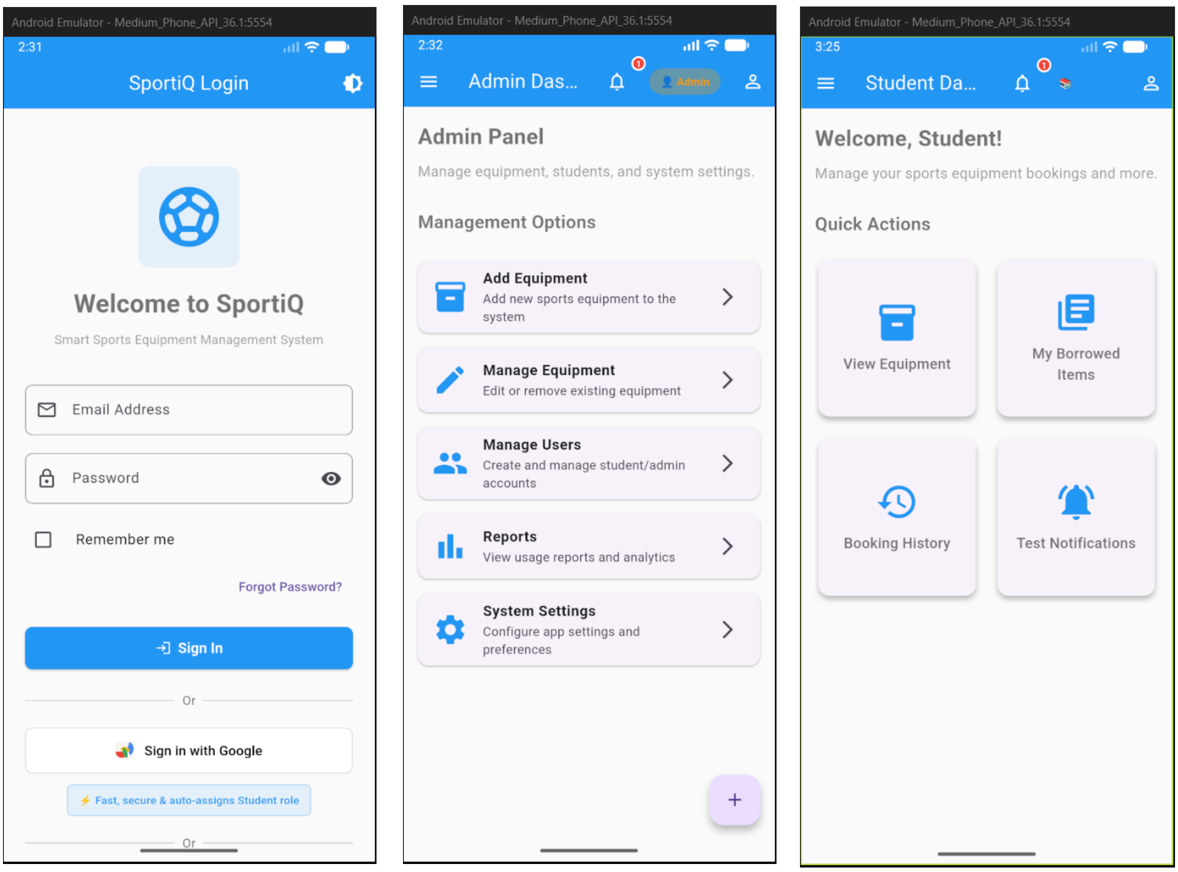


Fig 7.1 System Dashboard Interface

7.3 EQUIPMENT MANAGEMENT RESULT

The admin module allows efficient management of equipment inventory. Administrators can add, update, and remove equipment, and the system reflects changes in real-time.

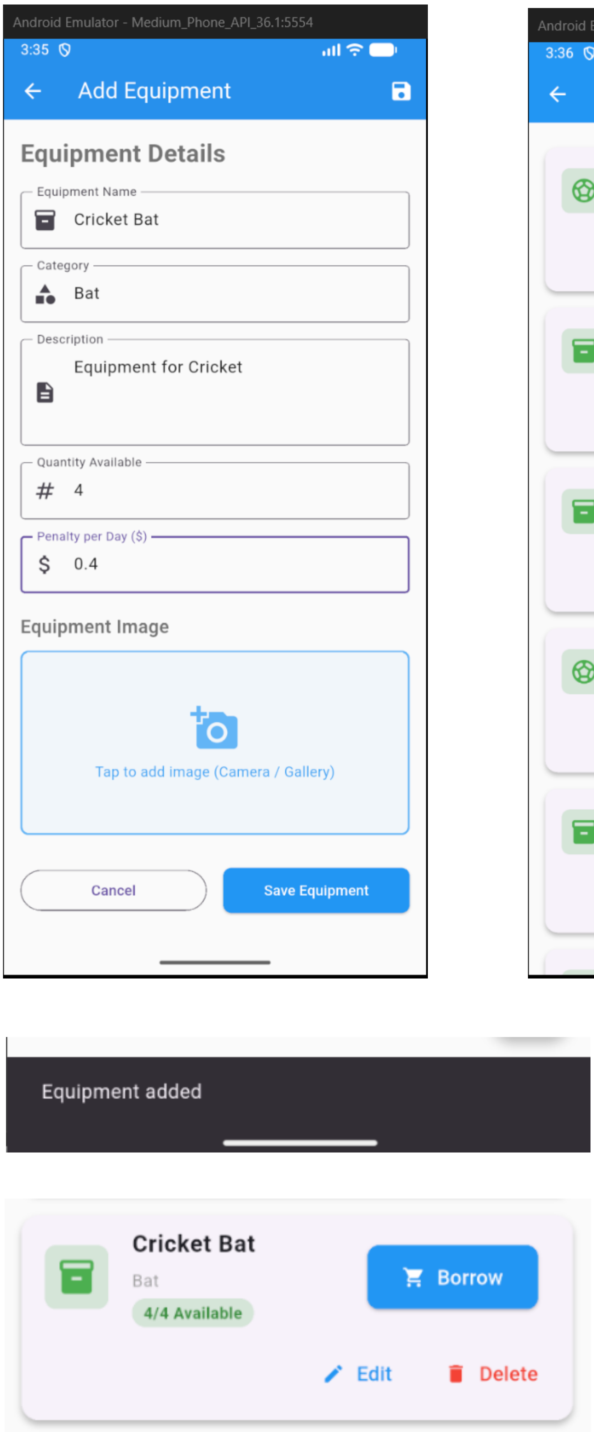


Fig 7.2 Equipment Management Module

7.4 BORROW RESULT

The borrowing process works smoothly with proper updates in the database. The system maintains accurate records of each transaction.

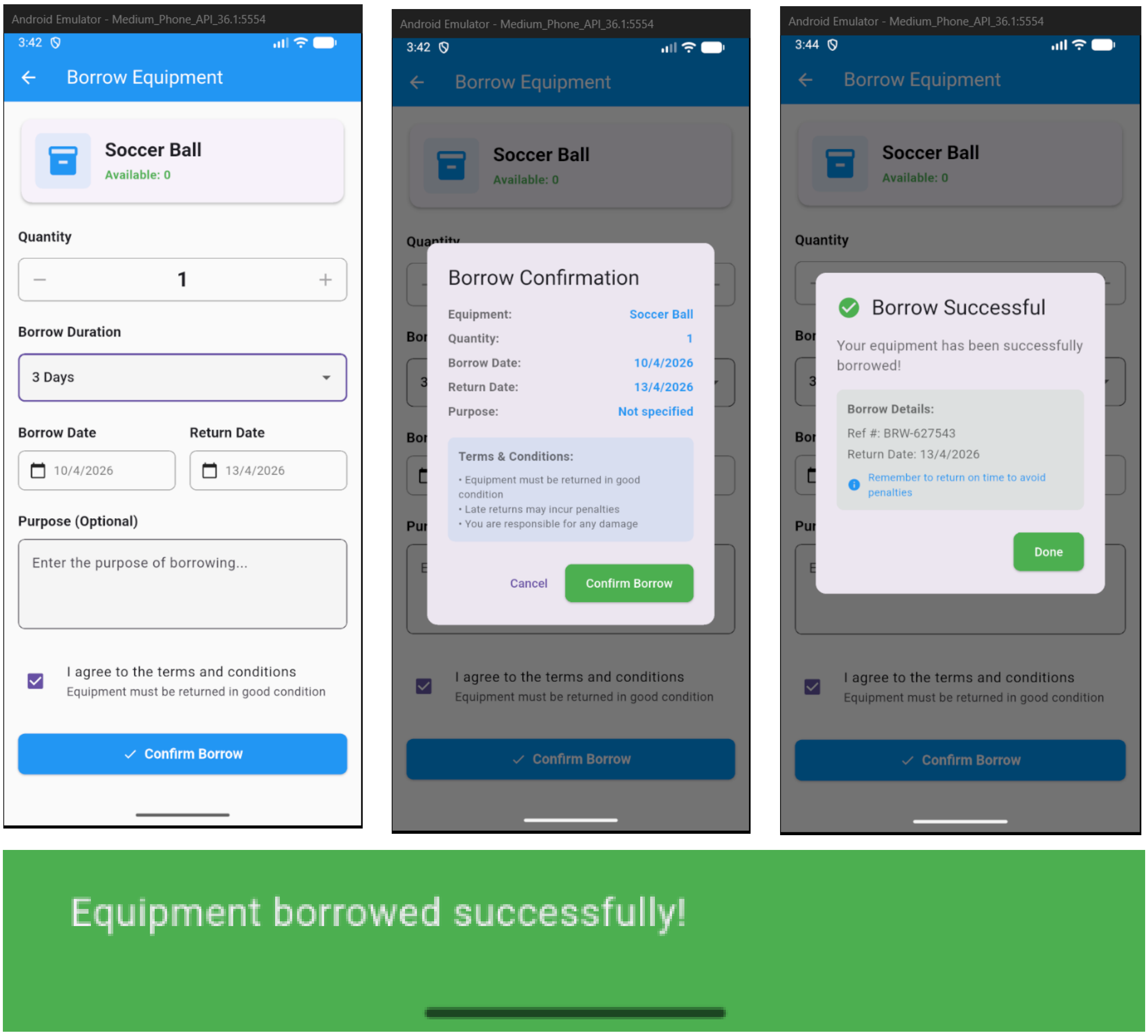
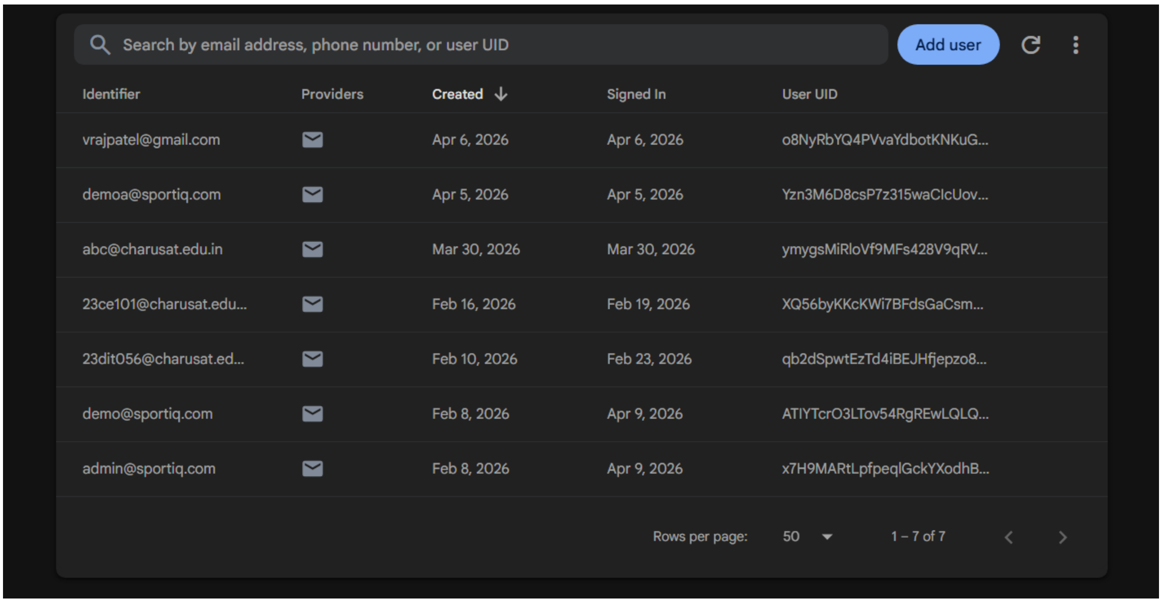


Fig 7.3 Borrow Workflow

7.5 DATABASE RESULT

The MongoDB database successfully stores and manages all records related to users, equipment, and rentals. Relationships between collections are maintained using ObjectId references.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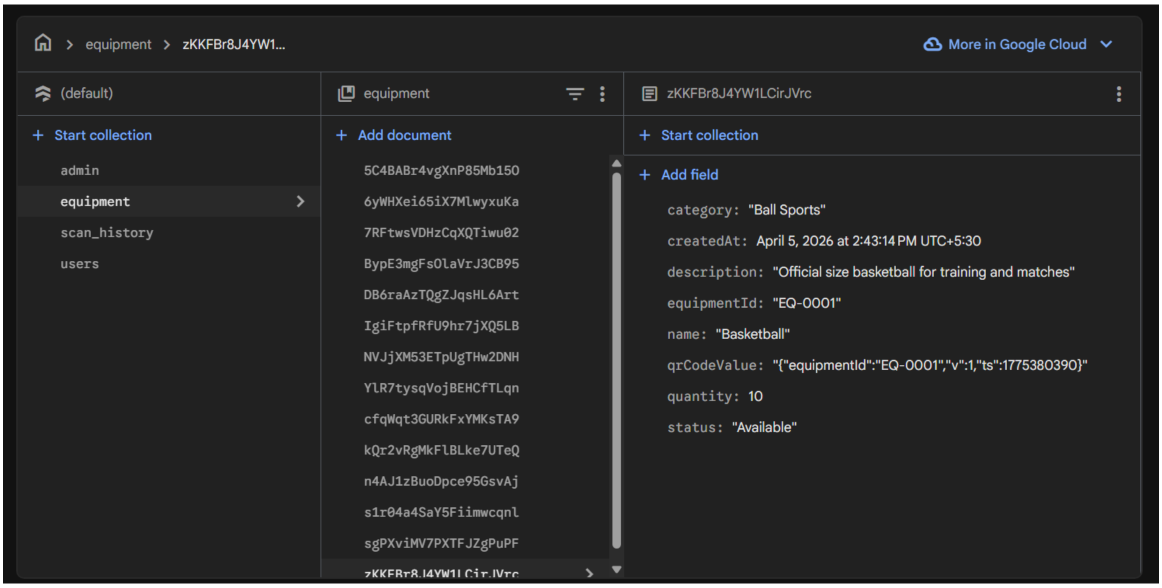
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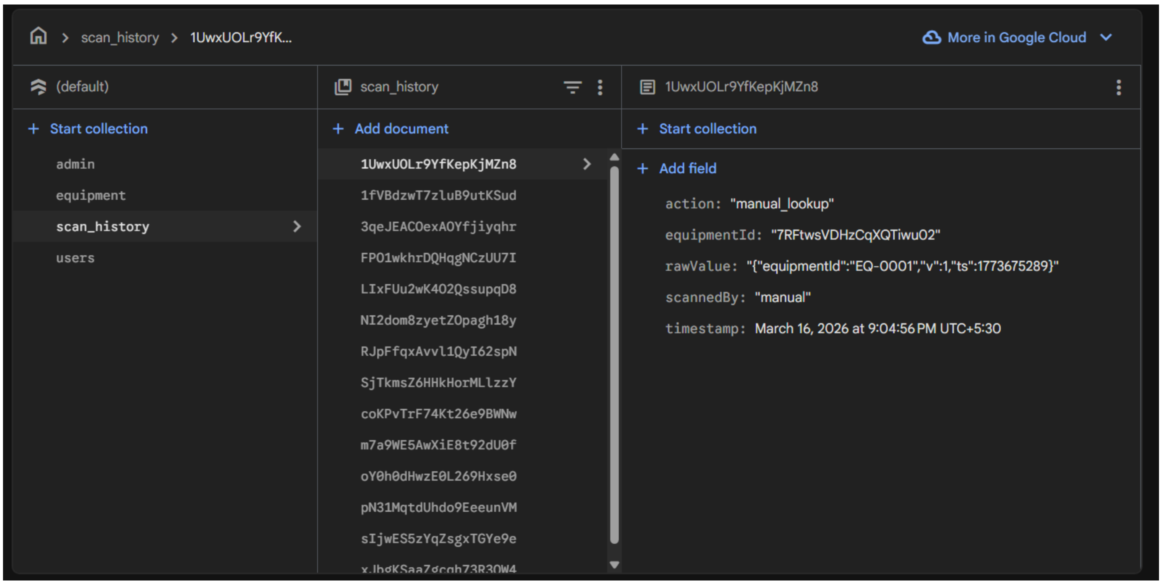
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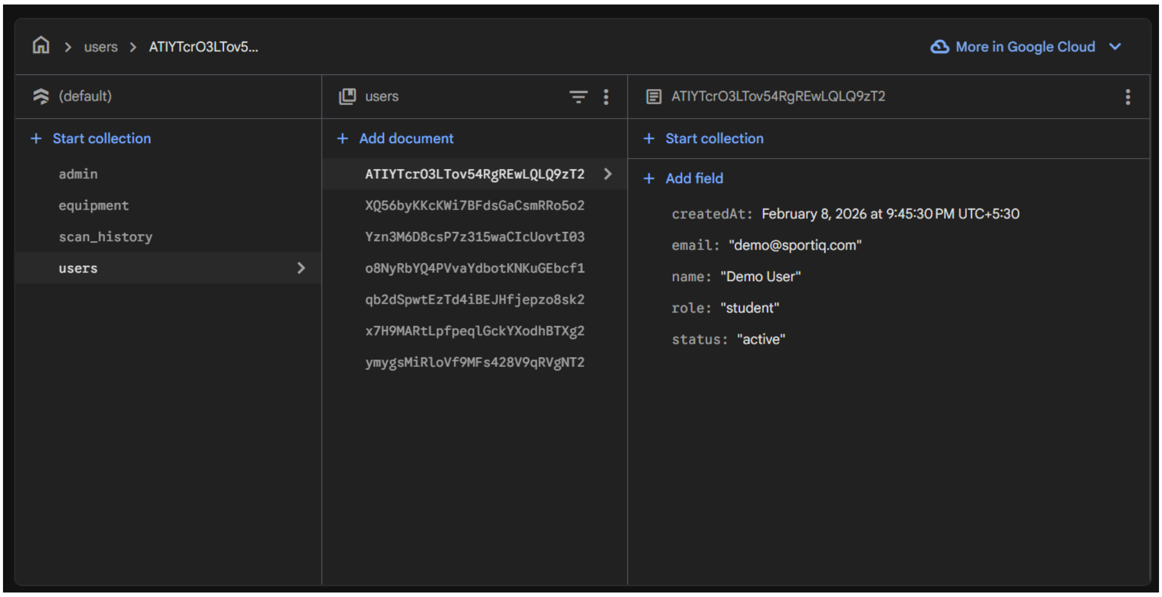

Fig 7.4 Database Records in MongoDB

7.6 API RESPONSE RESULT

The backend APIs function correctly and provide accurate responses for all operations such as fetching equipment data, creating rental records, and updating status.

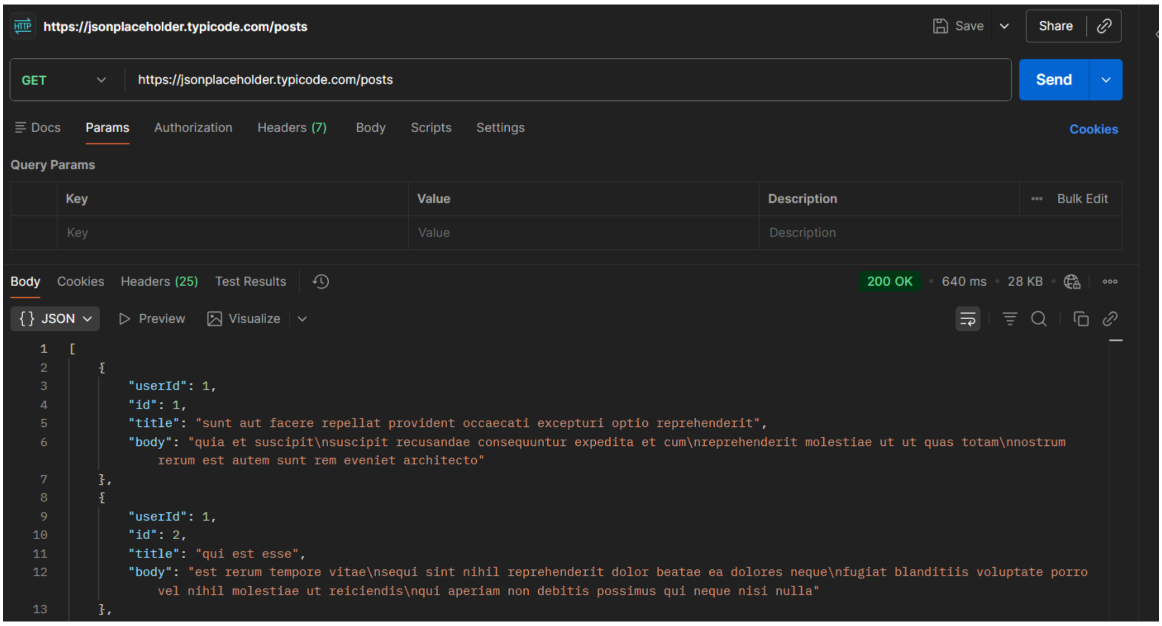

Fig 7.5 API Response Verification

7.7 PERFORMANCE RESULT

- The system provides fast response time
- Real-time updates are reflected accurately

- Data consistency is maintained across all modules

7.8 SUMMARY OF RESULTS

The SportiQ system successfully achieves its objectives by providing a reliable, efficient, and user-friendly solution for sports equipment management. The integration of frontend, backend, and database ensures smooth operation and improved overall performance.

CHAPTER 8: CHALLENGES FACED

8.1 INTRODUCTION

During the development of the SportiQ system, several challenges were encountered at different stages of the project. These challenges helped in gaining practical knowledge and improving problem-solving skills.

8.2 TECHNICAL CHALLENGES

8.2.1 DATABASE DESIGN COMPLEXITY

Designing a scalable and efficient MongoDB schema was challenging. Deciding between embedding and referencing required careful consideration to maintain data consistency and performance.

8.2.2 DATA CONSISTENCY

Ensuring synchronization between equipment availability and rental records was difficult. Proper updates were required during borrow and return operations to avoid inconsistencies.

8.2.3 API INTEGRATION

Integrating frontend with backend APIs required handling multiple requests and ensuring correct data flow. Debugging API responses and handling errors was challenging.

8.2.4 AUTHENTICATION HANDLING

Implementing secure authentication using Firebase and managing user roles (student/admin) required proper validation and backend support.

8.3 DEVELOPMENT CHALLENGES

- Managing coordination between frontend, backend, and database modules
- Debugging errors during development
- Handling real-time updates in the system

8.4 SOLUTIONS IMPLEMENTED

- Used **ObjectId references** to maintain proper relationships
- Applied validation rules to ensure correct data entry

- Tested APIs using Postman to debug issues
- Ensured proper update logic for borrow and return operations

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8.5 LEARNING OUTCOMES

- Improved understanding of database design and relationships
- Gained experience in backend API development
- Learned debugging and testing techniques
- Understood real-world system development challenges

CHAPTER 9: CONCLUSION AND FUTURE SCOPE

9.1 CONCLUSION

The SportiQ system successfully addresses the challenges associated with manual sports equipment management in educational institutions. By introducing a digital platform, the system improves efficiency, transparency, and accuracy in managing equipment.

The application provides real-time tracking of equipment availability, simplifies the borrowing and return process, and ensures accountability through automated monitoring and penalty calculation. The integration of Flutter, Node.js, MongoDB, and Firebase Authentication results in a robust and scalable system.

Overall, the project demonstrates a complete software development lifecycle, from problem identification and requirement analysis to system design, implementation, and testing. The system meets all the defined objectives and provides a practical solution for real-world use.

9.2 FUTURE SCOPE

The SportiQ system can be further enhanced with additional features such as:

- Push notifications for due dates and reminders
- Advanced analytics dashboard for usage trends
- Online penalty payment integration
- Feedback and rating system for equipment
- Multi-campus support for larger institutions
- Integration with IoT devices for automated tracking

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APPENDICES

A.1 INTRODUCTION

Appendices include additional supporting materials that are not part of the main report but help in understanding the implementation and functionality of the system.

A.2 SAMPLE API REQUESTS AND RESPONSES

Example:

- **GET /equipment** → Fetch all equipment
- **POST /borrow** → Create rental record
- **PUT /return** → Update return status

A.3 DATABASE COLLECTION STRUCTURE

Collections used:

- User
- Equipment
- Rental

Include:

- Sample document structure
- Fields and relationships

A.4 USER INTERFACE

- Login Screen
- Dashboard
- Equipment List
- Borrow Screen
- Return Screen

A.5 ADDITIONAL DETAILS

- Extra configurations
- Error handling logs
- Testing outputs